



# RAMCO INSTITUTE OF TECHNOLOGY

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## Department of Computer Science and Business Systems

Academic Year 2024–2025 (Even Semester)

**Degree, Semester & Branch:** IV Semester B.Tech CSBS

**Course Code & Title:** CS3492 Database Management Systems

**Name of the Faculty member:** Dr.S.Erana Veerappa Dinesh, ASP/CSBS

### Innovative Practice Description

- **Unit/Topic:** Unit 5 / Database Security - Access control based on privileges
- **Course Outcome:** CO5
- **Activity Chosen:** Peer Discussion Learning
- **Justification:**

Peer discussion helped to make the concepts of database security, especially access control based on privileges, more engaging and distinguishable from the regular syllabus. Through interactive learning, students gained in-depth understanding of how different privilege levels work to protect data from unauthorized access. Discussing real-world scenarios and case studies enabled them to relate theoretical concepts with practical applications. This approach not only enhanced their technical knowledge of roles, permissions, and user management in databases but also developed their non-technical skills such as teamwork, critical thinking, and effective communication. It encouraged students to approach security challenges collaboratively and understand the importance of maintaining data integrity and confidentiality in a professional environment.

- **Time Allotted for the Activity:** 20 minutes
- **Details of the Implementation:**
  - ✓ Individual topics related to access control and privilege management were given to the students prior to the start of the event for preparation purposes.
  - ✓ After the completion of the unit on Database Security, particularly focusing on Access Control Based on Privileges, the students participated in this activity.
  - ✓ Two Students of II Year CSBS actively shared their views, understanding, and examples related to the allotted topics during the session.
  - ✓ Finally, the faculty member consolidated the key points and insights discussed, summarizing the concepts of access control mechanisms, privilege levels, and their

importance in securing database systems.

• **CO – PO / PSO mapping:**

| CO  | PO1 | PO4 | PSO1 |
|-----|-----|-----|------|
| CO2 | 3   | 2   | 2    |

(1 – Low      2 – Moderate      3 – High)

**PO / PSO mapped:**

| Innovative practice                  | PO1                                                                                                                                                                | PO4                                                                                                                                                                                   | PSO1                                                                                                                                                                           |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | 3                                                                                                                                                                  | 2                                                                                                                                                                                     | 2                                                                                                                                                                              |
| <b>Justification for correlation</b> | Fundamental knowledge of database concepts and security mechanisms is required to understand and implement access control techniques using privileges effectively. | Research-based knowledge is applied to analyze real-time security threats and design secure database systems using appropriate access control strategies, including GRANT and REVOKE. | The student is able to apply domain-specific knowledge in designing and implementing secure, role-based access controls to safeguard data in real-world database applications. |

**Screenshot of the practice:**



*Fig.1. Peer Discussion by the student Subhash Selvam K*

**Reflective Critique:**

- The problem-solving capabilities of students will be improved.
- It is helpful for the students to understand the concept of access control based on privileges.

***Benefit of the practice:***

- Students can able to attend the question even in the questions are in indirect form.
- Students can able to explain the concepts without any confusion.
- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

***Challenges faced in implementation:***

- Some students feel that the discussion is tough due to their lack of interest.
- Students feel like a time-consuming activity.
- An equal contribution from all the students is not possible.

**References:**

<https://www.edsys.in/what-is-peer-teaching/>

<https://www.opencolleges.edu.au/informed/features/peer-teaching/>

<https://tilt.colostate.edu/TipsAndGuides/Tip/180>